

Basic Information

Term	Fall 2025
Course Title	Software Engineering
Course Code	CSE3203
Section	2
Credit Hours	3.0
Class Schedule	Monday and Wednesday: 08:00 am – 09:20 pm
Classroom link:	https://classroom.google.com/c/NzgwNjg3Mzc0MzI5?cjc=byvb74r2
Pre-requisite Course	CSE 3103
Department Offering the Course	Computer Science and Engineering
Faculty	Atanu Shuvam Roy
Contact Email	Atanu.shuvam@ulab.edu.bd
Contact Number	01937545261
Office and Location	PC319, Level 3, Building C, Permanent Campus
Counseling/Office Hour	Monday 10 AM – 12 PM Note: Also available by appointment (e.g. email) at other hours
Number of Lectures	24

1.0 Course Description (from syllabus)/Rational:

Most useful software products are large, complicated and built over a significant length of time. Such endeavors usually involve the participation of many different people of varied skill levels. Software Engineering is a discipline which is mainly interested in how we can build such complex systems in a systematic manner, so that a high-quality product is ensured. The discipline is necessarily large and involves a wide variety of topics. This course will concentrate mainly on Software Requirement Specification and Analysis, Design, Testing, Documentation and to a more limited extent - implementation. Students will be taught design principles, patterns and techniques. They will be challenged to design systems at a scale many of them are yet to encounter. They will be required to work in large teams and design and implement complete solutions as a part of their assessment. Over the semester, students will be expected to learn details about certain existing technologies, some of which they will also be required to master.

1.1 Course Objectives:

- Explain fundamental concepts of Software Engineering

- Apply software development methodologies to develop software solutions for real-life problems
- Formulate a testing strategy for a software quality assurance

1.2 Course Outcome (CO): (at the end of the course, students will be able to do:)

CO1	Explain fundamental concepts of Software Engineering	Cognitive / L2, Affective / L2
CO2	Apply software development methodologies to develop software solutions for real-life problems	Cognitive / L4, Affective / L3
CO3	Formulate a testing strategy for a software quality assurance	Cognitive / L5, Affective / L2

1.3 Teaching and Learning Activities (TLA)

TLA1	Interactive discussion using Online/multimedia or whiteboard.
TLA2	Interactive video and/or scenario-based presentation
TLA3	Case Study and group discussion
TLA4	Real-life project in a team to apply data structure knowledge

1.4 Course Delivery Plan (include Lab if any)

Week/Lesson (hour)	Discussion Topic & Book Reference	Student Activities during Online and Onsite and TLA	Assessment and Mapping CO
Wk 1 Lesson 1 & 2 (1.5x2 = 3.0)	Lesson-1: Importance of Software & Software Engineering, Essential attributes of Good Software Lesson-2: Project and Group Overview, Expectation setup, Core Development Technology study	Lesson-1 & 2: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	Background Preparation week
Wk 2 Lesson 3 & 4 (1.5x2 = 3.0)	Lesson-3: Software Process Model; Process Activities, Discussion on Software Process Models; Waterfall Model; V-Model Lesson-4: Iterative Models; Spiral Model; Incremental & Evolutionary Model	Lesson-3&4: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2
Wk 3 Lesson 5 & 6 (1.5x2=3.0)	Lesson-5: Review Discussion on Iterative Models; Agile Process Models. Lesson-6: Extreme Programming (XP); Scrum	Lesson-5&6: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2, TLA3	CO1, CO2
Wk 4 Lesson 7 & 8 (1.5x2=3.0)	Lesson-7: Quiz and Adapting a Process Model.	Lesson-7 & 8: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2

	Requirements Analysis Models; Functional and Nonfunctional requirements, Lesson-8: Require Specification, Validation, Scenario Based Models		
Wk 5 Lesson 9 & 10 (1.5x2=3.0)	Lesson-9: System Modeling: Context, Interaction, Structural, Lesson-10: Behavioral, Model Driven Architecture Mid Viva	Lesson-9 & 10: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2 Class Test#1 Interview
Wk 6 Lesson 11 & 12 (1.5x2=3.0)	Lesson 11: Requirements Analysis Models; Scenario Based Models Lesson 12: Review discussion for Mid Exam;	Lesson-11 & 12: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3
Wk 7	Midterm Exam Week Topics: Wk 3 – Wk 6		
Wk 8 Lesson 13 & 14 (1.5x2=3.0)	Lesson 13: OOD using UML, Design pattern, Implementation issues Lesson 14: Usage Scenario; Use Case Diagrams; Activity Diagrams	Lesson-13& 14: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3
Wk 9 Lesson 15 & 16 (1.5x2=3.0)	Lesson 15: Architectural design: Decisions, views, patterns and architecture. Lesson 16: Discussion on Architectural design, ER Diagram & Schema Tables	Lesson-15 & 16: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3 Class Test #2
Wk 10 Lesson 17 & 18 (1.5x2=3.0)	Lesson 17: Software Evaluation: Evaluation process, Legacy System, Software Maintenance Lesson 18: Quiz and Project Topic Selection	Lesson-17 & 18: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3
Wk 11 Lesson 19 & 20 (1.5x2=3.0)	Lesson 19: Discussion on System Dependability and Security Lesson 20: Discussion on System Security Semester Final Viva (Tentative)	Lesson-19 & 20: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3
Wk 12 Lesson 21 & 22 (1.5x2=3.0)	Lesson 21: Project Management: Risk management, Managing people Lesson 22: Project Planning, Plan driven development, agile planning, project scheduling Semester Final Viva (Tentative)	Lesson-21 & 22: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3 Class Test #3 Interview
Wk 13 Lesson 23 & 24 (1.5x2=3.0)	Lesson 23: Project Presentation and review for final exam Lesson 24: Project Presentation and review for final exam	Lesson-23& 24: Online/Onsite discussion; Review Feedback online; PPT, Video etc; TLA1, TLA2	CO1, CO2, CO3

Wk 14	Final Exam Week Topics: Wk 8, Wk 9, and Wk 10
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1.6 Text and Reference Materials

Text Book(s): Software Engineering, 9th Edition, By Ian Sommerville, Pearson

Reference Material/Book(s):

- (1) Software Engineering -- A Practitioner's Approach. Sixth Edition, By Roger S. Pressman, McGraw Hill
- (2) Gamma, E., 1995. *Design patterns: elements of reusable object-oriented software*. Pearson Education India
- (3) Chauhan, N., 2010. *Software Testing: Principles and Practices*. Oxford university press

1.7 Distribution of Marks for Assessment

- **Mid Semester Exam – 25%**
- **Final Semester Exam – 40%**
- **Class Attendance & Performance – 10%**
- **Course Project – 25%**

Grading Policy

Policy	Letter Grade	Grade Point	Assessments
95% and above	A+	4.00	Outstanding
85% to 94%	A	4.00	Superlative
80% to 84%	A-	3.80	Excellent
75% to 79%	B+	3.30	Very Good
70% to 74%	B	3.00	Good
65% to 69%	B-	2.80	Average
60% to 64%	C+	2.50	Below Average
55% to 59%	C	2.20	Passing
50% to 54%	D	1.50	Probationary
below 50%	F	0.00	Fail
--	I	0.00	Incomplete
--	W	0.00	Withdrawn
--	AW	0.00	Administrative Withdrawal

..... <i>Course Coordinator/ Teacher</i> Date: 25/05/2025	 <i>Head of the Department</i> Date: 25/05/2025

Appendix-1: Program outcomes

POs	Category	Program Outcomes
PO1	Engineering Knowledge	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.
PO3	Design/Development of Solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.
PO4	Investigations	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO5	Modern tool usage	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Environment sustainability and	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.
PO9	Individual work and teamwork	Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.
PO10	Communication	Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.
PO12	Life Long Learning	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.